

DevOps Curriculum — Tools & Concepts

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1 DevOps & AWS Learning Hub

A structured path from zero to advanced, plus two handbook series:

Resource	Target size	Status
DevOps Curriculum	Reference list	Complete
AWS Curriculum	Reference list	Complete
DevOps Handbook	~1,200 pages	TOC + Part I-II written; Parts III-XII outlined
AWS Handbook	~800 pages	TOC + Part I-II written; Parts III-X outlined

1.1 Download PDFs

Pre-built PDFs are in the [pdf/](#) folder:

PDF	Description
devops-handbook.pdf	DevOps handbook — 11 chapters + full TOC
aws-handbook.pdf	AWS handbook — 11 chapters + full TOC
devops-curriculum.pdf	DevOps tools & concepts reference
aws-curriculum.pdf	AWS services reference
devops-aws-complete-guide.pdf	All content in one PDF

Rebuild PDFs after editing markdown:

```
./scripts/build-pdfs.sh
```

Requires `pandoc` and `wkhtmltopdf`.

- 1. **Start with the curricula** — they list every tool, concept, and AWS service in learning order.
- 2. **Follow the handbook TOCs** — each chapter has a page budget; work through sequentially.
- 3. **Expand on demand** — ask an AI assistant or instructor to flesh out any chapter number from the TOC.

1.2 Suggested study pace

Phase	Duration (self-paced)	Focus
Foundations	4-8 weeks	Linux, Git, networking, scripting
Core DevOps	8-12 weeks	CI/CD, containers, IaC, cloud basics
Intermediate	8-12 weeks	Kubernetes, observability, security
Advanced	12+ weeks	SRE, platform engineering, multi-

Advanced	12+ weeks	engineering, multi-cloud, FinOps
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1.3 Page convention

Each handbook “page” $\approx$ 400 words of instructional content + diagrams/commands. Labs and exercises add extra pages in the TOC estimates.

2 DevOps Curriculum: Tools & Concepts (Beginner → Advanced)

Complete reference of what to learn, in recommended order. Items marked **(adv)** are advanced topics.

2.1 Level 0 — Prerequisites (before DevOps)

2.1.1 Concepts

- Software development lifecycle (SDLC)
- Agile, Scrum, Kanban
- Waterfall vs iterative delivery
- Client-server architecture
- APIs (REST, GraphQL basics)
- Databases (SQL vs NoSQL overview)
- Environment tiers: dev, staging, production

2.1.2 Tools & Skills

Category	Tools / Skills
Operating systems	Linux (Ubuntu, RHEL), Windows Server basics
Shell	Bash, Zsh, PowerShell basics
Text editing	Vim, Nano, VS Code
Networking	TCP/IP, DNS, HTTP/HTTPS, TLS, ports, firewalls
Version control	Git (clone, branch, merge, rebase, cherry-pick)
Scripting	Python, Bash, Go (pick one primary)
Documentation	Markdown, diagrams (draw.io, Mermaid)

2.2 Level 1 — DevOps Foundations

2.2.1 Concepts

- What is DevOps? Culture, CALMS (Culture, Automation, Lean, Measurement, Sharing)
- DevOps vs SysAdmin vs SRE vs Platform Engineering
- Continuous Integration (CI) vs Continuous Delivery (CD) vs Continuous Deployment
- Build artifacts, immutable infrastructure
- Configuration vs secrets
- The Twelve-Factor App
- DevSecOps introduction
- GitOps introduction

2.2.2 Tools

Category	Tools
Git hosting	GitHub, GitLab, Bitbucket, Azure DevOps Repos
SSH & keys	OpenSSH, ssh-agent, key pairs

Package managers	apt, yum/dnf, Homebrew
Build tools	Make, Maven, Gradle, npm/yarn/pnpm
Virtualization	VirtualBox, VMware, Hyper-V
Cloud CLI basics	AWS CLI, gcloud, az

2.3 Level 2 — Infrastructure & Automation

2.3.1 Concepts

- Infrastructure as Code (IaC) principles
- Idempotency, declarative vs imperative
- Server provisioning & bootstrapping
- Configuration management
- Immutable vs mutable infrastructure
- Golden images / AMIs
- Inventory & state management

2.3.2 Tools

Category	Tools
IaC — Terraform	Terraform, OpenTofu, Terragrunt
IaC — Cloud-native	AWS CloudFormation, Azure Bicep/ARM, Google Deployment Manager, Pulumi, CDK (AWS/Azure/GCP)
Configuration mgmt	Ansible, Chef, Puppet, SaltStack
Image building	Packer
Secrets (intro)	HashiCorp Vault, AWS Secrets Manager, SOPS
Cloud VMs	EC2, Azure VMs, GCE

2.4 Level 3 — Containers & Orchestration

2.4.1 Concepts

- Containers vs VMs
- Images, layers, registries
- Dockerfile best practices
- Container networking & storage
- Orchestration primitives (Pod, Service, Deployment)
- Service mesh concepts
- Sidecar pattern
- Helm charts & package management

2.4.2 Tools

Category	Tools
Runtime	Docker, containerd, CRI-O, Podman
Compose / local	Docker Compose, Dev Containers

Registries	Docker Hub, ECR, GCR, ACR, Harbor, Artifactory
Orchestration	Kubernetes (K8s), Amazon ECS, Amazon EKS, GKE, AKS
K8s tooling	kubectl, k9s, Lens, Helm, Kustomize
Service mesh	Istio, Linkerd, Consul Connect
Local K8s	minikube, kind, k3s, Docker Desktop K8s

2.5 Level 4 — CI/CD Pipelines

2.5.1 Concepts

- Pipeline stages: source → build → test → scan → deploy
- Branching strategies (GitFlow, trunk-based)
- Artifact versioning & promotion
- Blue/green, canary, rolling deployments
- Feature flags
- Pipeline as Code
- DORA metrics (deployment frequency, lead time, MTTR, change failure rate)

2.5.2 Tools

Category	Tools
CI/CD platforms	Jenkins, GitHub Actions, GitLab CI, CircleCI, Travis CI, Azure Pipelines, AWS CodePipeline, Tekton, Argo CD (GitOps)
Build agents	Jenkins agents, GitHub-hosted/self-hosted runners
Artifact repos	Nexus, Artifactory, GitHub Packages
GitOps	Argo CD, Flux, Fleet
Deployment	Spinnaker, Harness, Octopus Deploy

2.6 Level 5 — Cloud Platforms (cross-cutting)

2.6.1 Concepts

- Shared responsibility model
- Regions, AZs, edge locations
- IAM (users, roles, policies, least privilege)
- VPC networking (subnets, routing, NAT, peering)
- Load balancing (L4 vs L7)
- Object storage vs block vs file
- Managed vs self-managed services
- Well-Architected Framework pillars
- Cost optimization & tagging
- Multi-account / landing zone strategy

2.6.2 Tools (by provider)

Provider	Core services to learn
AWS	See curriculum-aws.md

Azure	Resource Manager, AAD, VNet, AKS, App Service
GCP	IAM, VPC, GKE, Cloud Run, Cloud Build

2.7 Level 6 — Observability & Reliability

2.7.1 Concepts

- Monitoring vs observability (metrics, logs, traces)
- SLI, SLO, SLA, error budgets
- Alerting philosophy (symptom vs cause)
- On-call, incident management, postmortems
- Chaos engineering
- Capacity planning
- Distributed tracing

2.7.2 Tools

Category	Tools
Metrics	Prometheus, Grafana, Datadog, New Relic, CloudWatch, Azure Monitor
Logging	ELK/EFK (Elasticsearch, Logstash/Fluentd, Kibana), Loki, Splunk, CloudWatch Logs
Tracing	Jaeger, Zipkin, OpenTelemetry, AWS X-Ray, Honeycomb
APM	Datadog APM, Dynatrace, AppDynamics
Incident mgmt	PagerDuty, Opsgenie, incident.io
Status	Statuspage
Chaos	Chaos Monkey, LitmusChaos, Gremlin

2.8 Level 7 — Security (DevSecOps)

2.8.1 Concepts

- Shift-left security
- OWASP Top 10
- Supply chain security (SBOM, signing)
- Zero trust networking
- CIS benchmarks
- Compliance frameworks (SOC 2, PCI-DSS, HIPAA overview)
- Threat modeling
- Principle of least privilege

2.8.2 Tools

Category	Tools
SAST	SonarQube, Semgrep, CodeQL, Checkmarx
DAST	OWASP ZAP, Burp Suite

Container scan	Trivy, Grype, Snyk, Clair
IaC scan	Checkov, tfsec, KICS, cfn-lint
Secrets scan	git-secrets, TruffleHog, Gitleaks
Policy as code	OPA, Gatekeeper, Kyverno, Sentinel
WAF / edge	AWS WAF, Cloudflare, ModSecurity
PKI / certs	cert-manager, Let's Encrypt, ACM
SIEM	Splunk, Elastic Security, AWS Security Hub

2.9 Level 8 — Data, Messaging & Databases (Ops view)

2.9.1 Concepts

- Backup & restore strategies (RPO/RTO)
- Replication, failover, DR
- Connection pooling
- Caching layers
- Message queues & event-driven architecture
- Database migrations in CI/CD

2.9.2 Tools

Category	Tools
RDBMS	PostgreSQL, MySQL, MariaDB, SQL Server
NoSQL	MongoDB, DynamoDB, Redis, Cassandra
Messaging	Kafka, RabbitMQ, SQS, SNS, Pub/Sub
Streaming	Kinesis, Flink, Spark Streaming
Backup	Velero (K8s), AWS Backup, pg_dump, Percona XtraBackup
Migrations	Flyway, Liquibase, Alembic

2.10 Level 9 — Platform Engineering & Internal Developer Platforms (adv)

2.10.1 Concepts

- Platform as a product
- Developer experience (DevEx)
- Self-service portals
- Golden paths & paved roads
- Service catalogs
- Backstage software templates

2.10.2 Tools

Category	Tools
IDP	Backstage, Port, Cortex
API gateways	Kong, Apigee, AWS API Gateway, Ambassador

Service discovery	Consul, Eureka, CoreDNS
Feature flags	LaunchDarkly, Unleash, Flagsmith

2.11 Level 10 — Advanced & Specialized (adv)

2.11.1 Concepts

- Site Reliability Engineering (SRE) — toil reduction, automation
- FinOps & cloud cost governance
- Multi-cloud & hybrid cloud
- Edge computing & CDN strategy
- WebAssembly in ops (adv)
- eBPF for observability (adv)
- ML/LLM ops (MLOps) pipelines

2.11.2 Tools

Category	Tools
FinOps	Kubecost, CloudHealth, AWS Cost Explorer, Infracost
CDN	CloudFront, Cloudflare, Fastly, Akamai
DNS	Route 53, Cloudflare DNS, ExternalDNS
Workflow	Airflow, Temporal, Step Functions
MLOps	MLflow, Kubeflow, SageMaker (AWS)
eBPF	Cilium, Pixie, Falco

2.12 Certifications (optional milestones)

Track	Certifications
Linux	LPIC, RHCSA/RHCE
Cloud	AWS SAA, AWS DVA, AWS SOA, AWS DOP, AWS SAP; Azure AZ-104, AZ-400; GCP PCA
Kubernetes	CKA, CKAD, CKS
Terraform	HashiCorp Terraform Associate
Security	CISSP (broad), AWS Security Specialty
SRE	Google SRE book + practice (no single cert)

2.13 Learning project ideas (apply everything)

1. **Static site pipeline** — S3 + CloudFront + GitHub Actions + Terraform
2. **Microservices on EKS** — Helm, Prometheus, Argo CD
3. **Full observability stack** — app + OpenTelemetry + Grafana stack
4. **Secure CI/CD** — SAST/DAST/SCA in pipeline, signed images
5. **Multi-env IaC** — dev/stage/prod with Terragrunt or Terraform workspaces

6. **Incident simulation** — chaos tests + runbooks + PagerDuty
7. **Internal developer portal** — Backstage with one golden-path service